PAPER_1387_KLEIN_GORDON_NEGATIVE_ENERGY

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PAPER_1387 — UQFF Resolution of the Klein-Gordon Negative-Energy Paradox (CLOSED — Dispatch Whitepaper)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: B — Foundational / Relativistic QM (CLOSED)
Date: June 16, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "klein_gordon_negative_energy"})`
Calculator surface: `calculate_paradox({"paradox": "klein_gordon_negative_energy"})`
Closure helper: `_l96_uqff_axiom_klein_gordon_negative_energy_closure()`

The Paradox

The Klein-Gordon equation admits both E > 0 and E < 0 solutions. The negative-energy branch is paradoxical in single-particle interpretation — the spectrum is unbounded below.

UQFF Closed Identity

```
E_neg branch = E_pos branch in t_neg CCW dual existence (PAPER_597)
```

PAPER_597 negative-time dual existence: E < 0 in the CW (forward-time) branch IS E > 0 in the CCW (t_neg, backward-time) branch. The two branches coexist; neither is unphysical.

Physical Interpretation

The negative-energy spectrum is not a problem; it is the **t_neg dual existence branch** of the same field. Quantum field theory's antiparticle interpretation is the EM-coupling-aware version of this UQFF resolution.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "klein_gordon_negative_energy"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_klein_gordon_negative_energy_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_klein_gordon_negative_energy_closure`
- Dispatch keys: `klein_gordon_negative_energy`, `klein_gordon`, `negative_energy_paradox`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).
